## Why Are My Indexes Not Being Used?

As you saw earlier, even when you've added an Index you want PostgreSQL to use, it may choose to not use it. You saw that this can depend on many factors. Let's look at some of those factors in more depth.

Queries should be as selective as possible, meaning they select as few rows as possible, for your database server to scale better and achieve a higher volume of queries. Most of the Active Record queries you're writing or running are in an online transaction processing (OLTP) Ruby on Rails web application context where short durations are important.

Besides very selective queries, consider the cardinality of the columns you're working with. The number of distinct values in a column is the "cardinality." High cardinality means there are a lot of distinct values.

Indexes are ideal for high cardinality columns and highly selectivity queries. When there is a table index for a column with low cardinality, or in other words, the column has very few distinct values such as two or three total possible values, PostgreSQL may prefer a Sequential Scan over an Index Scan. In that case the Index Scan and Heap Scan may have a higher total cost compared with a Heap Scan only.

In the next section you'll begin to consider the definitions of your indexes. Indexes can cover single columns and multiple columns. Read on to learn more.

## Using Single Column And Multiple Column Indexes

*Multicolumn Indexes*[3] are indexes in PostgreSQL that include multiple columns in their definition. Terminology like "composite" or "compound" indexes from other relational databases are synonyms for Multicolumn Indexes in PostgreSQL.

Open the Rideshare database with psql. Run `VACUUM ANALZYE` without listing any tables to run it for all tables. Use the Index Maintenance query from the PostgreSQL wiki https://wiki.postgresql.org/wiki/Index_Maintenance to display information about all indexes for Rideshare grouped by tables.

Multicolumn Indexes are considered a specialized index. Most Indexes in Rideshare are single column indexes.

3. https://www.postgresql.org/docs/current/indexes-multicolumn.html